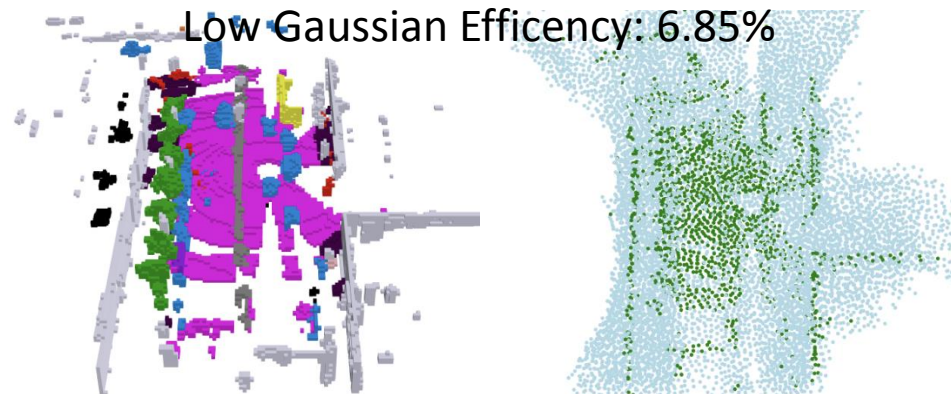
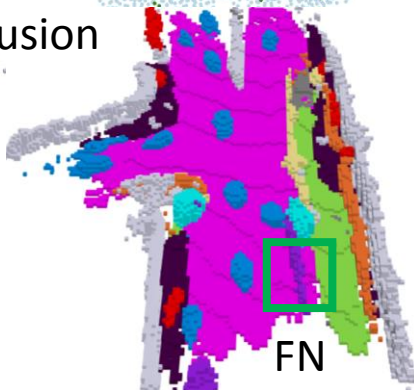
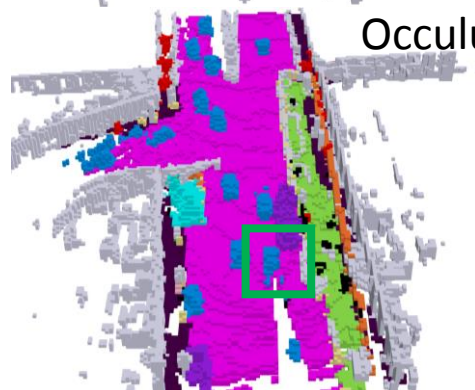


Low Gaussian Efficiency: 6.85%



Occulusion

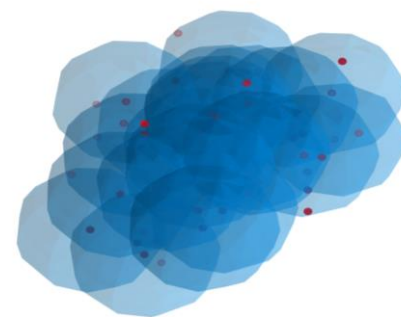


FN

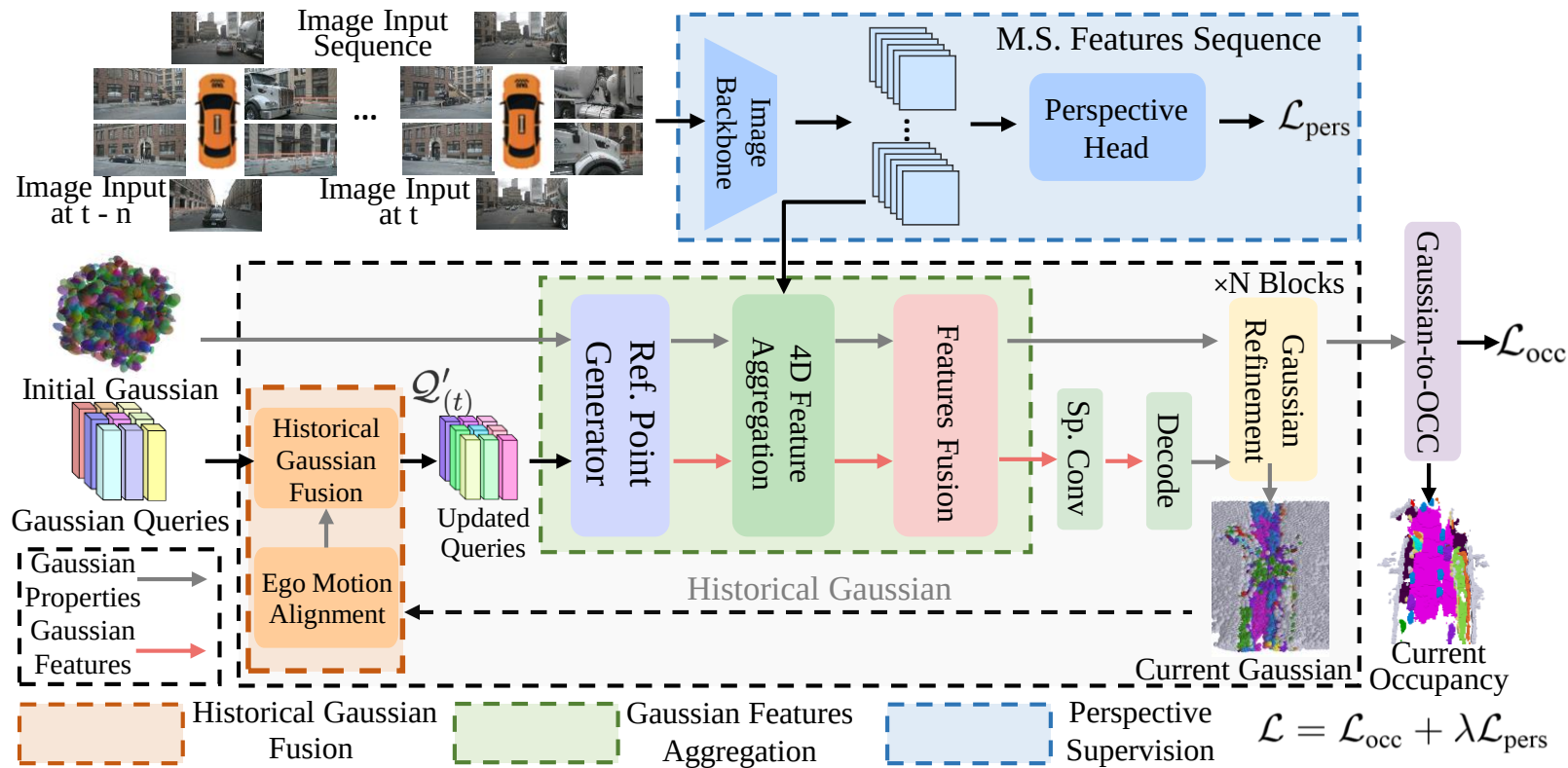
Sparse Sampling



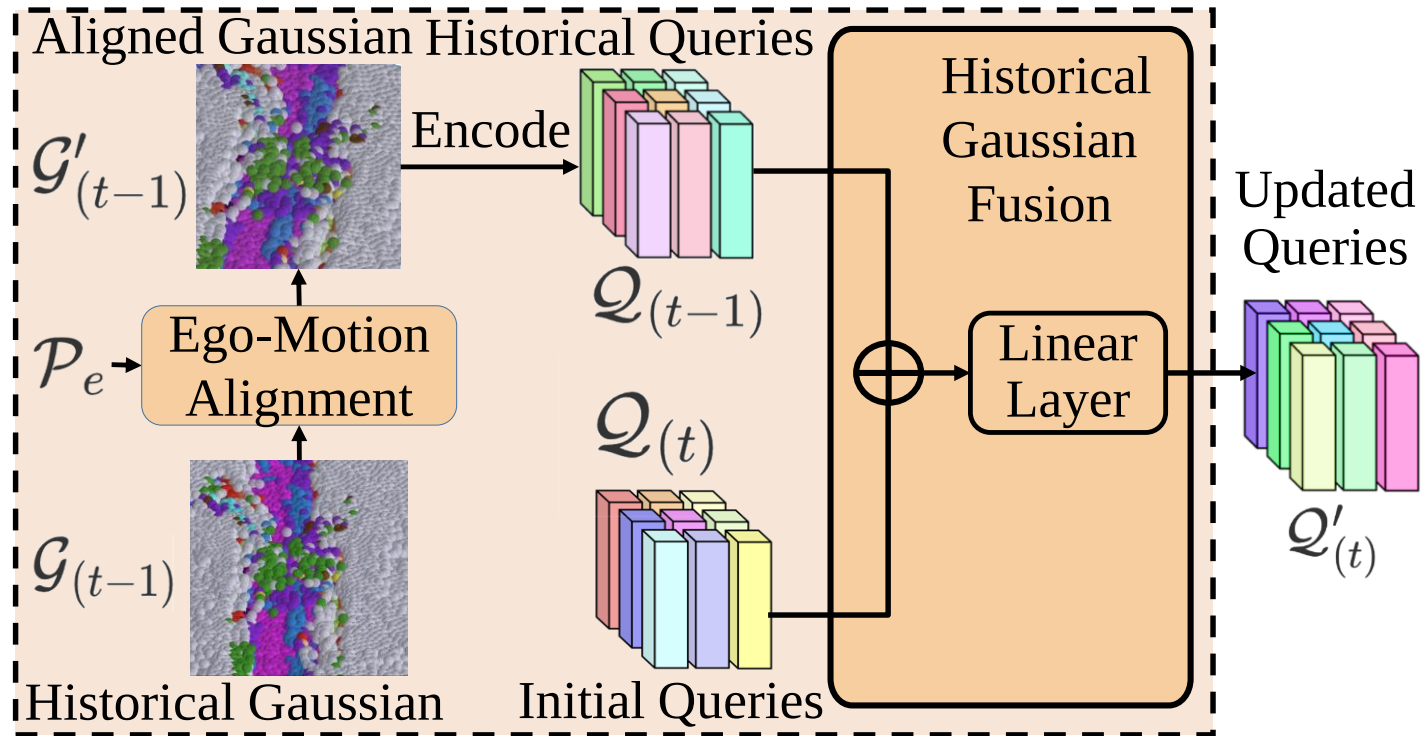
GrondTruth

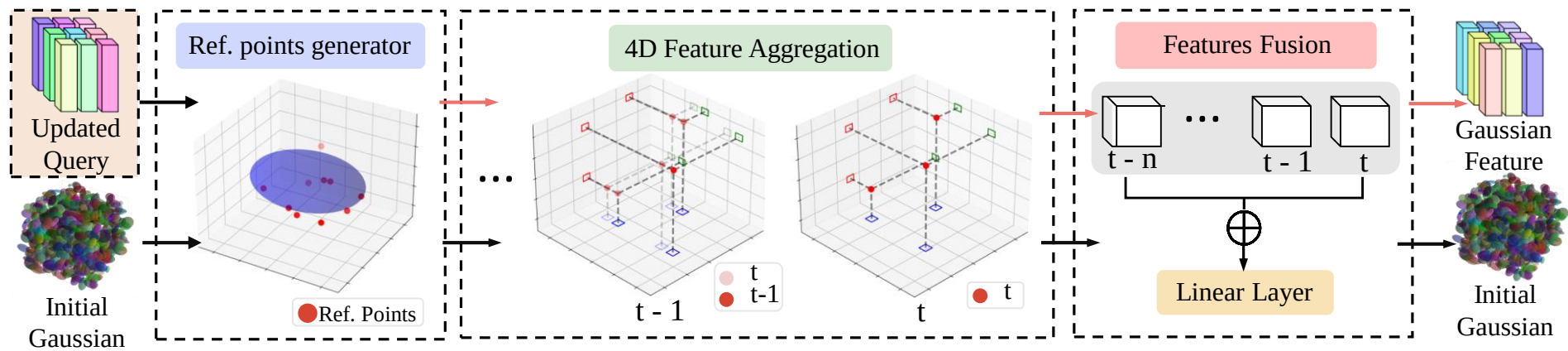


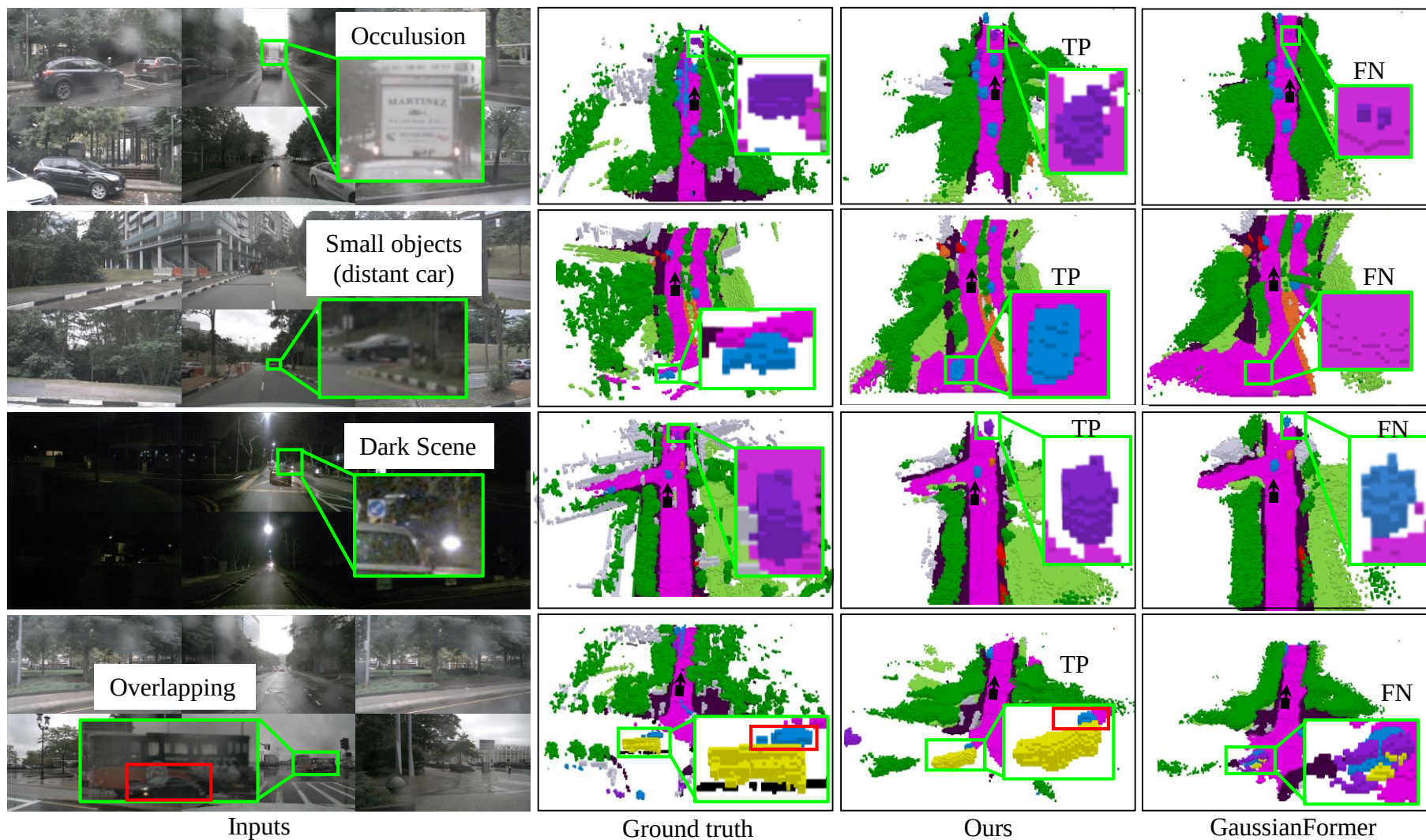
GaussianFormer

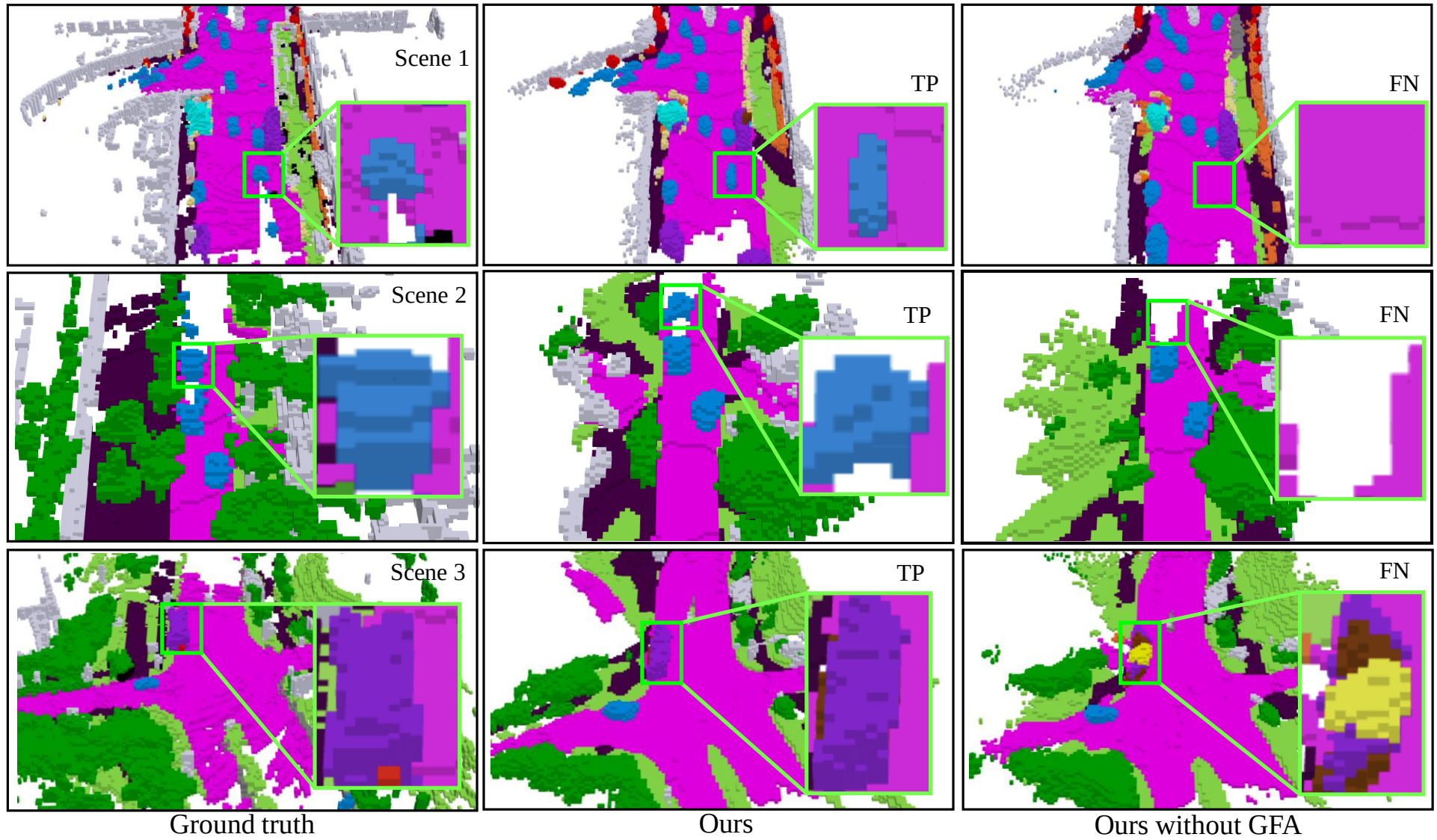


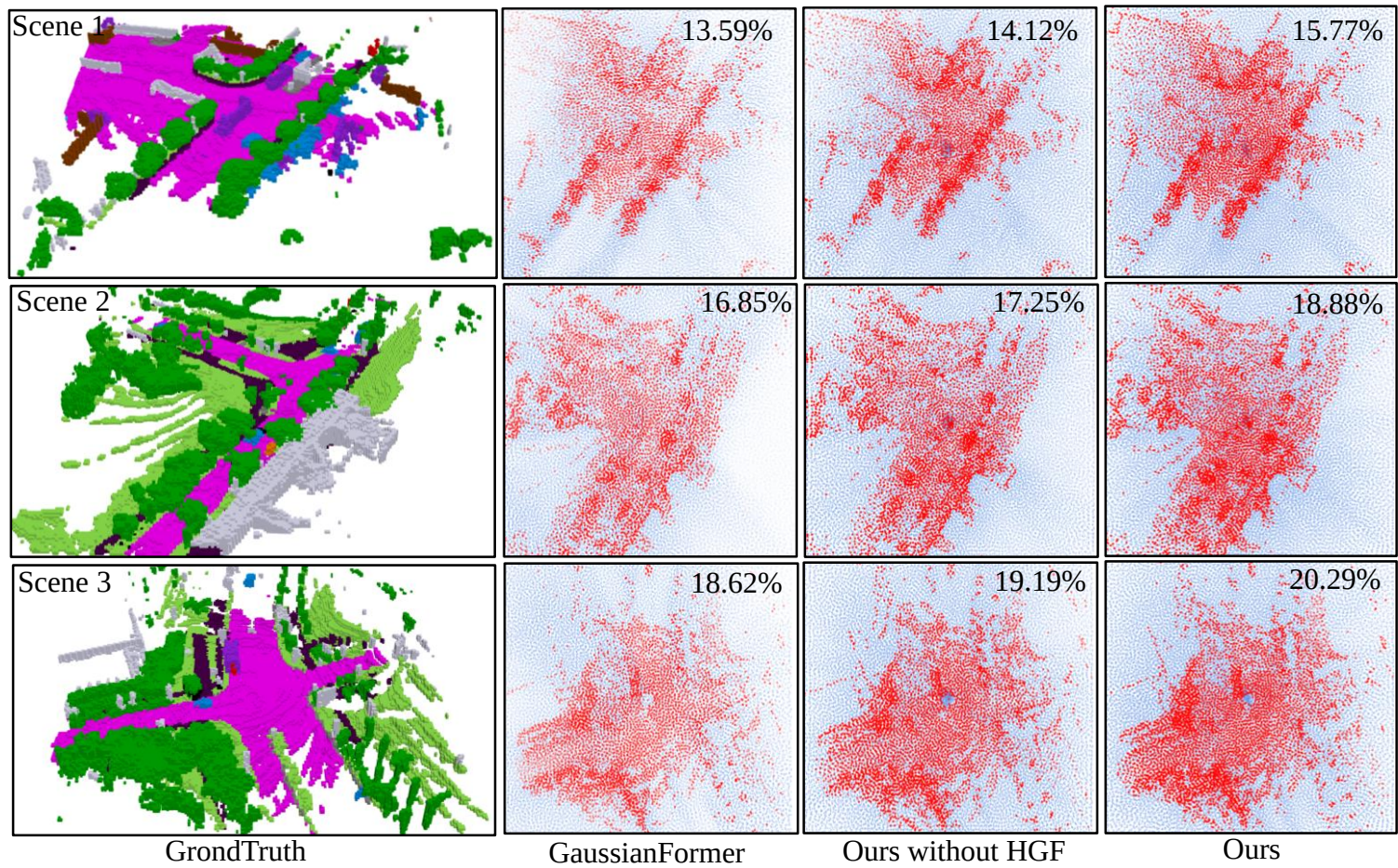
The framework of our Gaussian4DOCC semantic occupancy prediction contains three main parts: 1) historical Gaussian fusion, 2) Gaussian feature aggregation, and 3) perspective supervision.

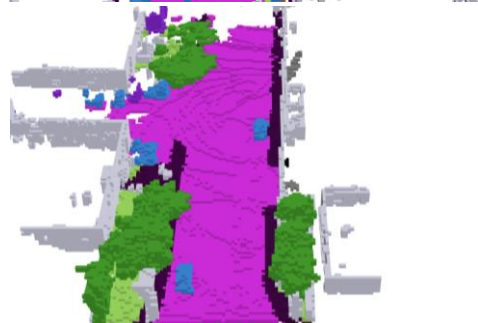
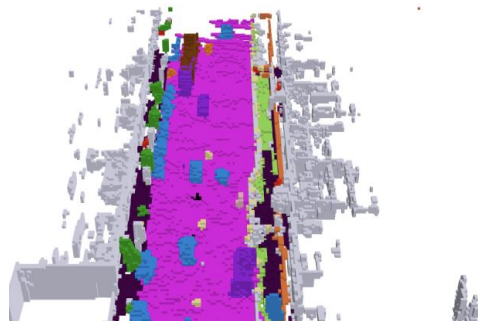








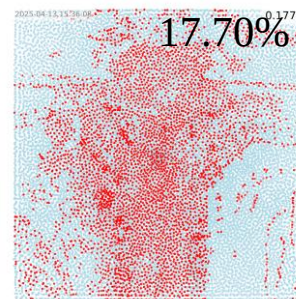




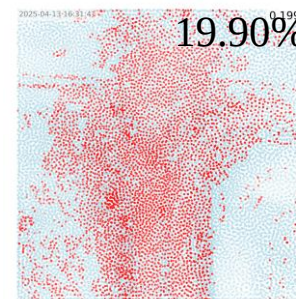
GrondTruth



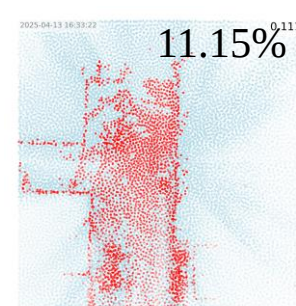
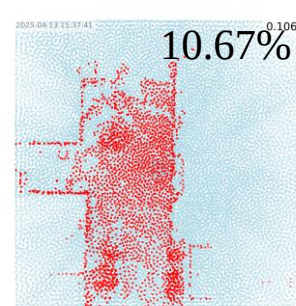
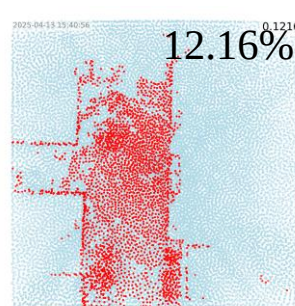
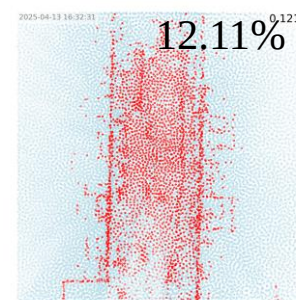
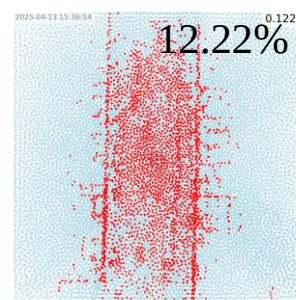
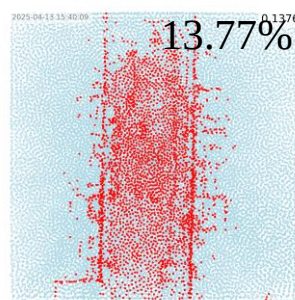
Ours

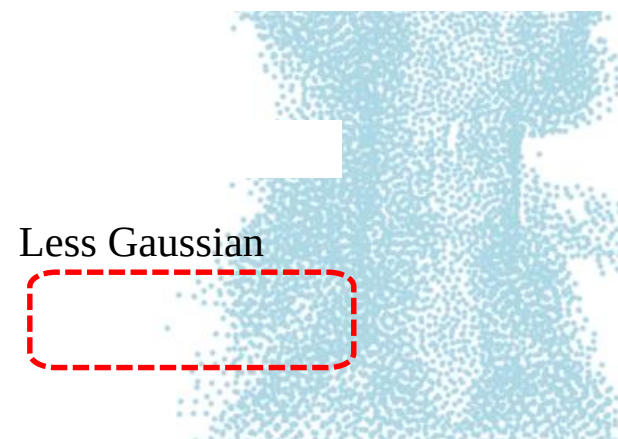
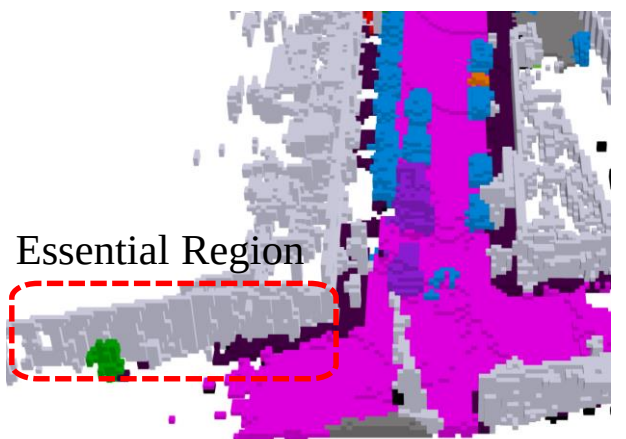
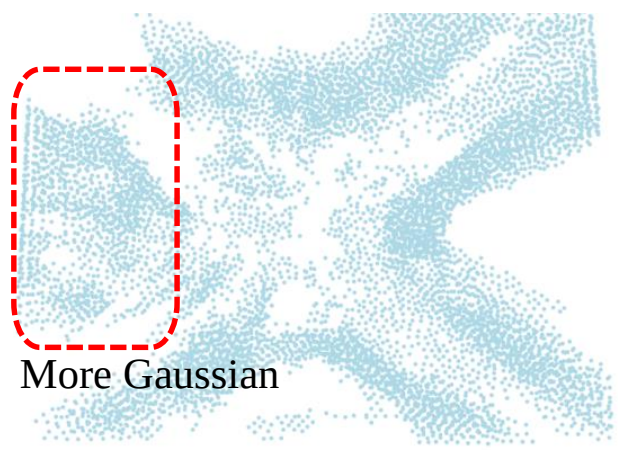
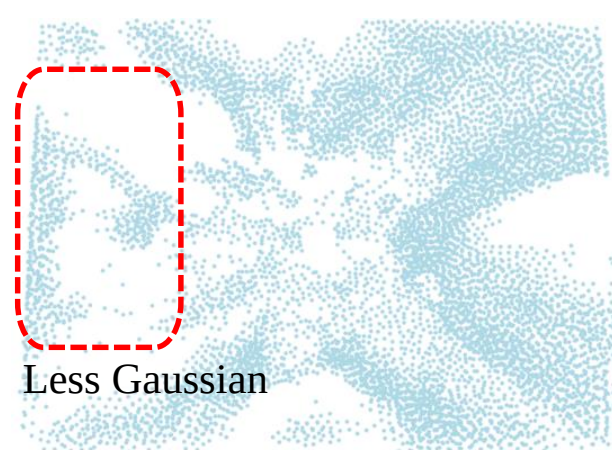
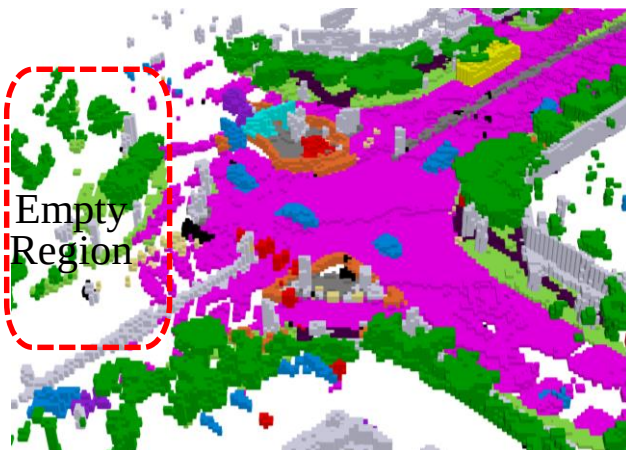
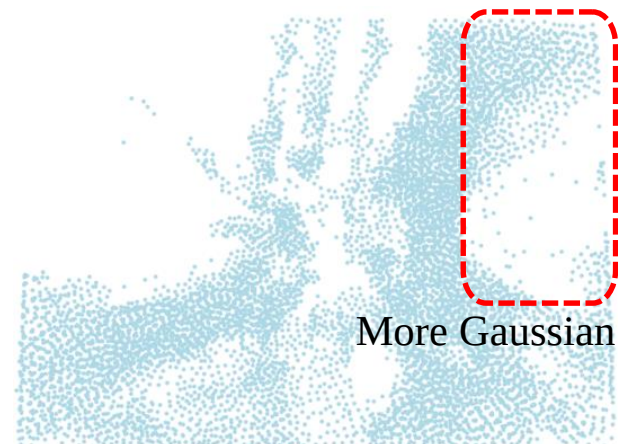
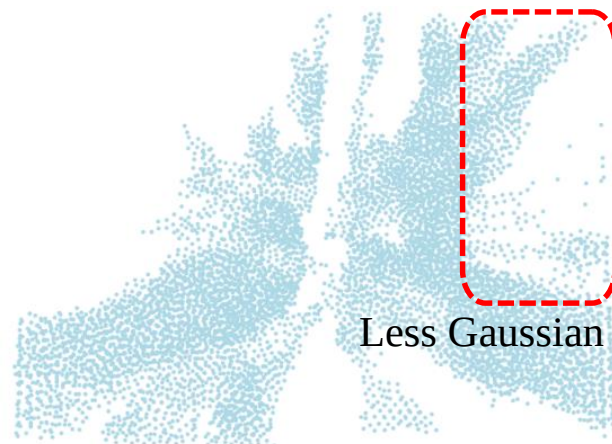
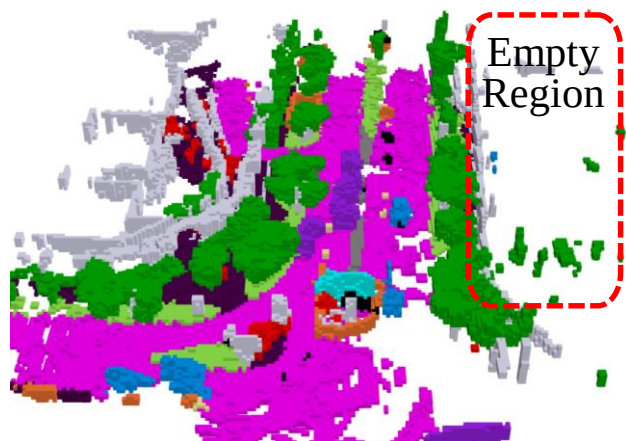


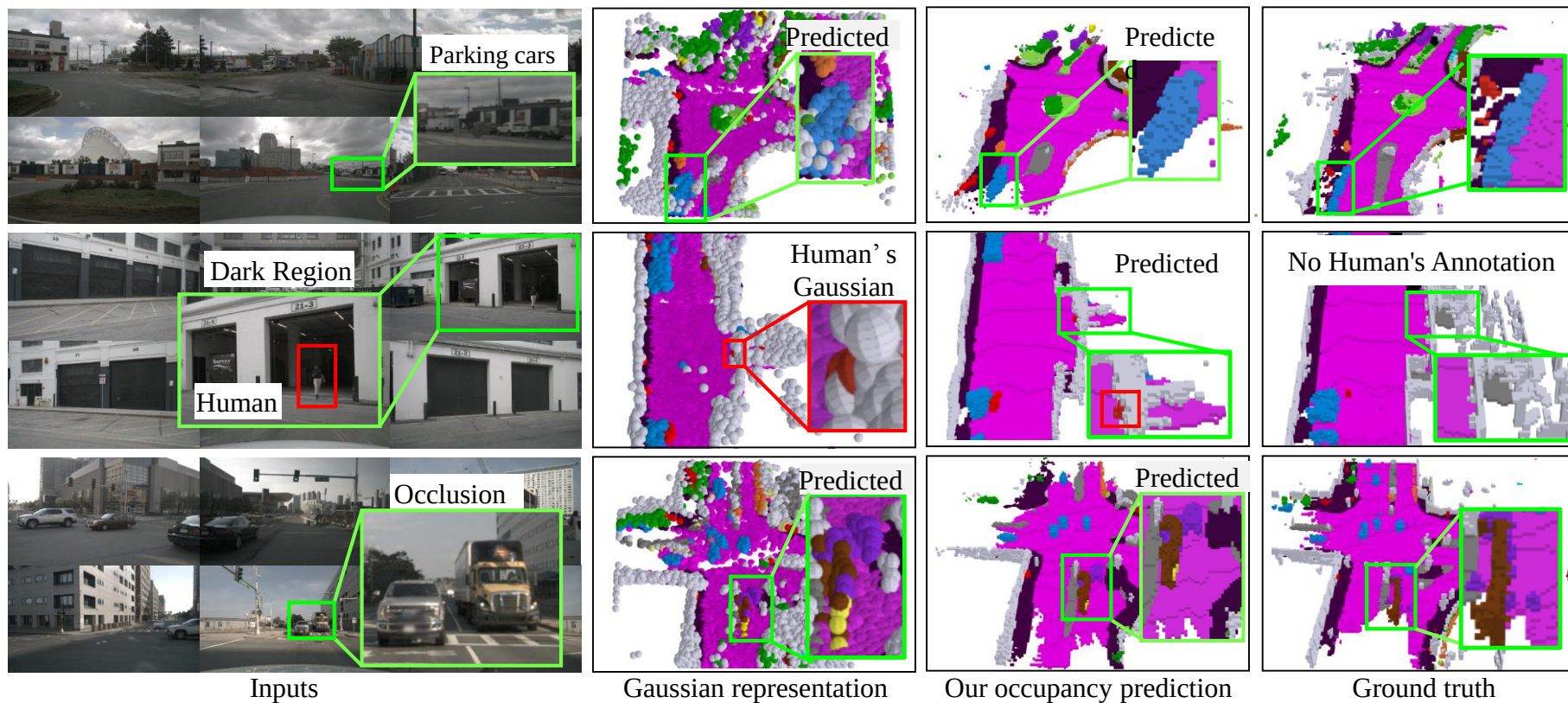
Ours without HGF

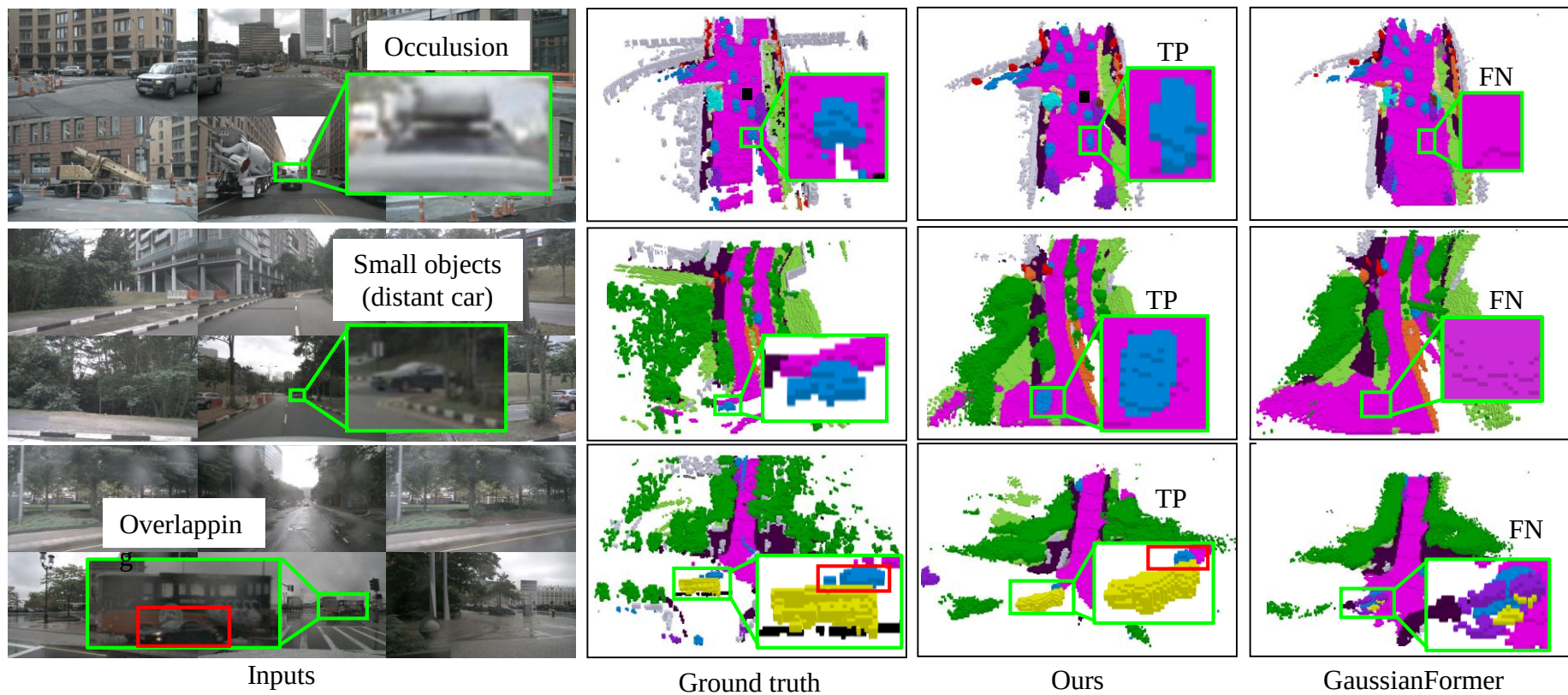


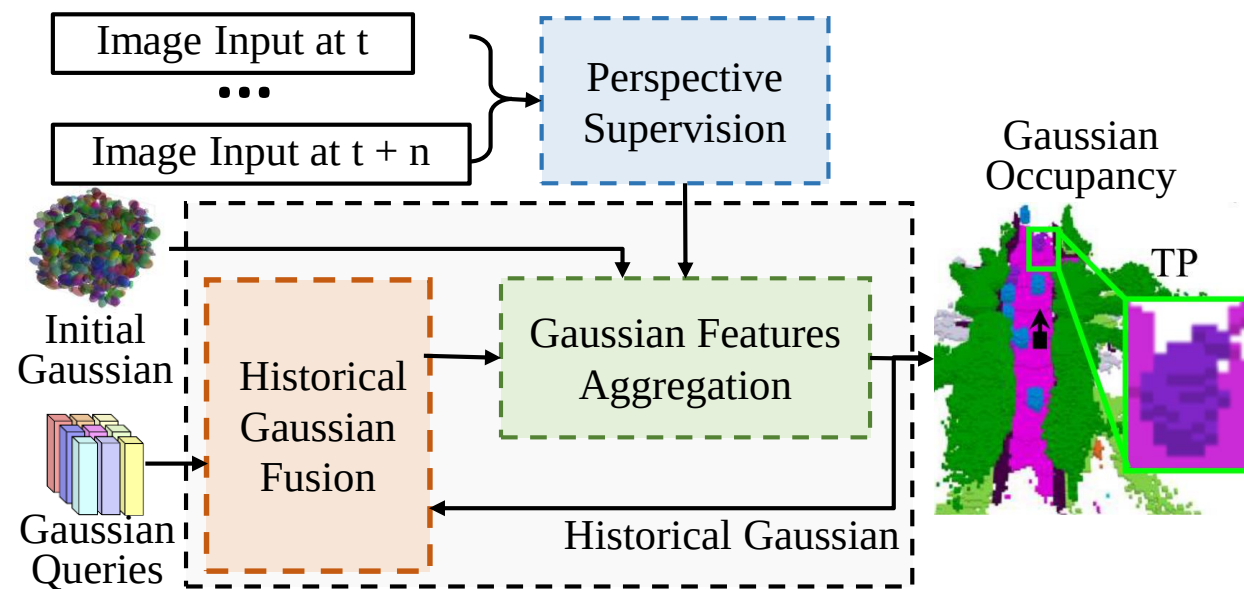
GaussianFormer











Our Gaussian4DOCC contains three contributions:

- 1) historical Gaussian fusion, 2) Gaussian feature aggregation,
- 3) perspective supervision

Low-angle sun glare during sunset often results in image overexposure and lane detection failure

对向远光灯

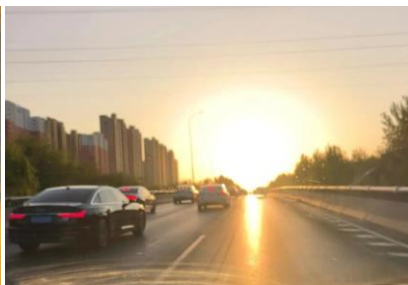
夕阳直射

剧烈光照变化

大雨 恶劣天气 大雾



Oncoming headlight glare



Low-angle sun glare



Heavy Rain



Dense Fog



Black hole effect



White hole effect



Heavy Snow



Dust Storm

隧道黑洞效应

隧道白洞效应

大雪

沙尘暴

The sudden illumination changes at tunnel portals cause the **black hole effect**(at the entrance) and the **white hole effect**(at the exit)